INCH-POUND

The documentation and process conversion measures necessary to comply with this document shall be completed by 6 January 2021.

MIL-PRF-19500/569B
6 October 2020
SUPERSEDING
MIL-PRF-19500/569A
19 August 2015

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, FIELD EFFECT, N-CHANNEL, SILICON,
CASE MOUNT THROUGH-HOLE MOUNT PACKAGE,
TYPES 2N6966, 2N6967, 2N6968, 2N6969,
QUALITY LEVELS JAN, JANTX, JANTXV, AND JANS

Inactive for new design after 7 June 1999.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for N-channel, enhancement-mode, MOSFET, power transistor intended for use in high density power switching applications. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device type.

1.2 Package outlines. The device package outline is a diamond base, flange mount TO-204AE (modified TO-3) in accordance with figure 1.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25 °C.

Type	P_T (1) T_C = +25 °C	P_T T_A = +25 °C	$R_{\theta JC}$ (2)	V_{DS}	V_{DG}	V_{GS}	I_{D1} (3) (4) T_C = +25 °C	I_{D2} (3) T_C = +100 °C	I_S	I_{DM}	T_J and T_{STG}
	W	W	°C /W	V dc	V dc	V dc	A dc	A dc	A dc	A(pk)	°C
2N6966	70	5	1.8	100	100	±20	15	13	15	±60	-55 to +150
2N6967	70	5	1.8	200	200	±20	13	8	13	±50	
2N6968	70	5	1.8	400	400	±20	7.5	5	7.5	±30	
2N6969	70	5	1.8	500	500	±20	6	4	6.0	±24	

(1) Derate linearly 0.56 W/°C for T_C > +25 °C.
(2) See figure 2 for thermal impedance curve.
(3) The following formula derives the maximum theoretical I_D limit. I_D is limited by package and internal wires and may be limited by pin diameter:

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \text{ X } (r_{DS(on)}) \text{ at } T_{JM}}}$$

(4) I_D = 4 x I_{D1} as calculated in note 2.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25 °C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 V I_D = 1 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = 0.25 mA		Max I_{DSS1} V_{GS} = 0 V	Max $r_{DS(on)1}$ (1) V_{GS} = 10 V dc, $I_D = I_{D2}$	
				V_{DS} = 80 percent of rated V_{DS}	T_J = +25 °C at I_{D1}	T_J = +150 °C at I_{D2}
	V dc	V dc Min	Max	µA dc	ohms	ohms
2N6966	100	2.0	4.0	25	0.085	0.13
2N6967	200	2.0	4.0	25	0.18	0.27
2N6968	400	2.0	4.0	25	1.0	1.04
2N6969	500	2.0	4.0	25	1.5	1.62

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV" and "JANS".

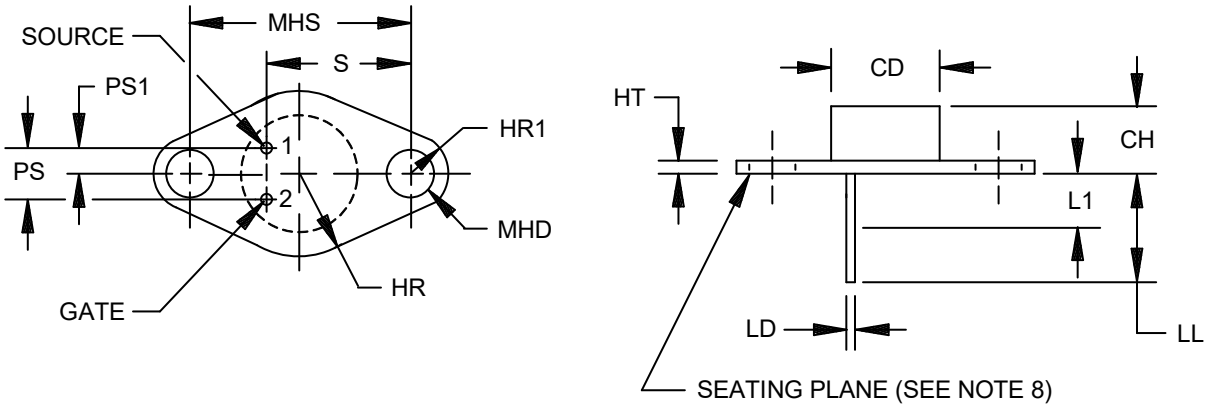
1.5.2 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The transistors of this specification sheet are identified by the first number and letter symbols "2N".

1.5.2.2 Second number symbols. The second number symbols for the transistor covered by this specification sheet are as follows: "6966", "6967", "6968", and "6969".

1.5.3 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

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Symbol	Dimensions				Notes	Symbol	Dimensions				Notes
	Inches		Millimeters				Inches		Millimeters		
	Min	Max	Min	Max			Min	Max	Min	Max	
CD		.875		22.22	3	LL	.440	.500	11.18	12.70	6
CH	.250	.450	6.35	11.43		L1		.050		1.27	6
HR	.495	.525	12.58	13.33		MHD	.151	.165	3.84	4.19	4
HR1	.131	.188	3.33	4.77	4	MHS	1.177	1.197	29.90	30.40	
HT	.050	.075	1.27	1.91	5	PS	.420	.440	10.67	11.18	7
LD	.057	.063	0.97	1.60	4, 6	PS1	.205	.225	5.21	5.72	7
						S	.655	.675	16.64	17.15	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Pin 1 is the Source, pin 2 is the Gate. The Drain shall be electrically connected to the case.
3. Body contour is optional within zone defined by dimension CD.
4. Two places.
5. Dimension HT does not include sealing flanges.
6. Dimension LD applies between dimensions L1 and LL. Lead diameter shall not exceed twice dimension LD within dimension L1.
7. These dimensions should be measured at points .050 inch (1.27 mm) +.005 inch (0.13 mm) to -.000 inch (0.00 mm) below seating plane. When gauge is not used, measurement will be made at the seating plane.
8. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
9. In accordance with ASME Y14.5M, diameters are equivalent to Øx symbology.

FIGURE 1. Physical dimensions for diamond base, flange mount TO-204AE package.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

nC	nano coulomb.
I_{AS}	Rated avalanche current, non-repetitive.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (similar to TO-204AE).

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction is not permitted to meet the requirements of this specification.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

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3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3 herein.

3.8 Electrostatic discharge sensitive (ESDS) protection. The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements of MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, R ≤ 100 kΩ, whenever bias voltage is to be applied drain to source.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (quality levels JANS, JANTX, and JANTXV only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (1) (2)	Measurement	
	JANS level	JANTX and JANTXV levels
(3)	Gate stress test (see 4.3.1)	Gate stress test (see 4.3.1)
(3)	Unclamped inductive switching (see 4.3.2) optional	Unclamped inductive switching (see 4.3.2) optional
(3) 3c	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
9	I_{GSSF1}, I_{GSSR1}, I_{DSS1}	Not applicable
10	Method 1042 of MIL-STD-750, test condition B	Method 1042 of MIL-STD-750, test condition B
11	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$, subgroup 2 of table I herein: ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater.	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$, subgroup 2 of table I herein.
12	Method 1042 of MIL-STD-750, test condition A, t = 240 hours	Method 1042 of MIL-STD-750, test condition A; or t = 48 hours minimum at +175 °C min
13	Subgroups 2 and 3 of table I herein; ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(th)1}$ = ±20 percent of initial value.	Subgroup 2 of table I herein; ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(th)1}$ = ±20 percent of initial value.

(1) At the end of the test program, I_{GSSF1}, I_{GSSR1} and I_{DSS1} are measured.
(2) An out-of-family program to characterize I_{GSSF1}, I_{GSSR1}, I_{DSS1} and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime after temperature cycling, screen 3a. Quality levels JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Gate stress test. Apply V_{GS} = 30 V minimum for t = 250 µs minimum.

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4.3.2 Single pulse unclamped inductive switching. The test for single pulse unclamped inductive switching shall be performed in accordance with method 3470 of MIL-STD-750. The following details shall apply:

a. Peak current, I_D.....Rated I_{D1}.

b. Peak gate voltage, V_{GS}.....10 V.

c. Gate to source resistor, R_{GS}.....25 Ω ≤ R_{GS} ≤ 200 Ω.

d. Initial case temperature T_C.....+25 °C +10 °C, -5 °C.

e. Inductance.....100 µH minimum.

f. Supply voltage V_{DD}.....50 V.

g. Number of pulses to be applied .....1 pulse minimum.

4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW}, t_{MD} (and V_H where appropriate). (See figure 2 herein.) See table II, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (for quality level JANS) and table E-VIB (for quality levels JAN, JANTX, and JANTXV) of MIL-PRF-19500, and as follows.

4.4.2.1 Quality level JANS (table E-VIA of MIL-PRF-19500).

Subgroup	Method	Conditions
B3	1051	Test condition G.
B4	1042	Test condition D. The heating cycle shall be 1 minute minimum.
B5	1042	Test condition A, V_{DS} = rated T_A = +175 °C, t = 120 hours minimum. Read and record $V_{(BR)DSS}$ (pre and post at 1 mA = I_D. Read and record I_{DSS} (pre and post). Deltas for $V_{(BR)DSS}$ shall not exceed 10 percent and I_{DSS} shall not exceed 25 µA.
B5	1042	Test condition B, V_{GS} = rated, T_A = +175 °C, t = 24 hours.
B5	2037	Test condition D.

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4.4.2.2 Quality level JAN, JANTX and JANTXV (table E-VIB of MIL-PRF-19500).

Subgroup	Method	Conditions
B2	1051	Test condition G.
B3	1042	Test condition D. The heating cycle shall be 1 minute minimum.
B3	2037	Test condition D. All internal bond wires for each device shall be pulled separately.
B4	2075	See 3.4.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	1056	Test condition A.
C2	2036	Test condition A; weight = 10 lbs. (4.54 kg), t = 15 s.
C5	3161	See 4.3.3.
C6	1042	Test condition D. The heating cycle shall be 1 minute minimum.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal response 2/	3161	See 4.3.3	$Z_{\theta JC}$			°C/W
Breakdown voltage, drain to source	3407	Condition C, V_{GS} = 0 V dc, I_D = 1 mA dc,	$V_{(BR)DSS}$			V dc
2N6966				100		
2N6967				200		
2N6968				400		
2N6969				500		
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 0.25 mA dc	$V_{GS(th)1}$	2.0	4.0	V dc
Gate current	3411	Bias condition C; V_{DS} = 0, V_{GS} = +20 V dc	I_{GSSF1}		±100	nA dc
Gate current	3411	Bias condition C; V_{DS} = 0, V_{GS} = -20 V dc	I_{GSSR1}		±100	nA dc
Drain current	3413	Bias condition C; V_{GS} = 0 V dc V_{DS} = 80 percent of rated V_{DS}; bias condition C	I_{DSS1}		25	µA dc
Static drain to source on-state resistance	3421	Condition A, V_{GS} = 10 V dc, pulsed (see 4.5.1), I_D = rated I_{D2} (see 1.3)	$r_{DS(on)1}$			ohms
2N6966					0.085	
2N6967					0.18	
2N6968					0.55	
2N6969					0.85	
Static drain to source on-state resistance	3421	Condition A, V_{GS} = 10 V dc, pulsed (see 4.5.1); I_D = rated I_{D1} (see 1.3)	$V_{DS(on)}$			V
2N6966					1.70	
2N6967					2.40	
2N6968					4.13	
2N6969					5.10	

See footnotes at end of table.

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TABLE I. Group A inspection – Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 − Continued						
Forward voltage (source drain diode)	4011	Condition C, $I_D = I_{D1}$, V_{GS} = 0 V	V_{SD}			V
2N6966				.85	2.5	
2N6967				.8	2.0	
2N6968				.8	2.0	
2N6969				.8	2.0	
Forward transconductance	3475	Pulsed (see 4.5.1)	g_{FS}			S
2N6966				6	18	
2N6967				6	18	
2N6968				4	12	
2N6969				4	12	
Subgroup 3						
High temperature operation:		$T_C = T_J$ = +125 °C				
Gate current	3411	Bias condition C, V_{GS} = ±20 V dc, V_{DS} = 0 V dc	I_{GSS2}		±200	nA dc
Drain current	3413	Bias condition C, V_{DS} = 80 percent of rated V_{DS}, V_{GS} = 0 V dc	I_{DSS2}		0.25	mA dc
2N6966						
2N6967						
2N6968						
2N6969						
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc pulsed (see 4.5.1); I_D = rated I_{D2}	$r_{DS(on)2}$			ohms
2N6966					0.13	
2N6967					0.27	
2N6968					1.04	
2N6969					1.62	
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc	$V_{GS(th)2}$	1.0		V dc

See footnotes at end of table.

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TABLE I. Group A inspection – Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 − Continued						
Low temperature operation:		$T_C = T_J = -55$ °C				
Gate to source voltage threshold	3403	$V_{DS} \geq V_{GS}$, $I_D = 0.25$ mA dc	$V_{GS(th)3}$		5.0	V dc
Subgroup 4						
Switching time test	3472	I_D = rated I_{D2} (see 1.3); V_{GS} = 10 V dc, Gate drive impedance = 7.5 Ω				
Turn-on delay time			$t_{d(on)}$			ns
2N6966		V_{DD} = 34 V dc			30	
2N6967		V_{DD} = 75 V dc			30	
2N6968		V_{DD} = 175 V dc			35	
2N6969		V_{DD} = 200 V dc			35	
Rise time			t_r			ns
2N6966		V_{DD} = 34 V dc			60	
2N6967		V_{DD} = 75 V dc			60	
2N6968		V_{DD} = 175 V dc			30	
2N6969		V_{DD} = 200 V dc			30	
Turn-off delay time			$t_{d(off)}$			ns
2N6966		V_{DD} = 34 V dc			80	
2N6967		V_{DD} = 75 V dc			80	
2N6968		V_{DD} = 175 V dc			90	
2N6969		V_{DD} = 200 V dc			90	
Fall time			tf			ns
2N6966		V_{DD} = 34 V dc			30	
2N6967		V_{DD} = 75 V dc			60	
2N6968		V_{DD} = 175 V dc			35	
2N6969		V_{DD} = 200 V dc			30	
Subgroup 5						
Safe operating area test	3474	See figure 3; V_{DS} = 80 percent of rated V_{DS}; tp = 10 ms				
Electrical measurements		See table I, subgroup 2				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition A and B				
Test 1 Minimum off-state gate charge			$Q_{g(th)}$	2.7	8.0	nC
Test 2 On-state gate charge			$Q_{g(on)}$	30	77	nC
Test 3 Maximum on-state gate charge			$Q_{gm(on)}$	51	130	nC
Test 4 Gate plateau voltage charge			V_{GP}	4	8	V dc
Test 5 Gate to source charge			Q_{gs}	4.6	13	nC
Test 6 Gate to drain charge			Q_{gd}	13	40	nC
Reverse recovery time	3473	di/dt = 100 A/μs, $V_{DD} \leq 30$ V, $I_D = I_{D1}$	t_{rr}			ns
2N6966					400	
2N6967					650	
2N6968					800	
2N6969					1000	

1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group B, subgroups 3 and 4 (JANS).
Group C, subgroup 6.
Group E, subgroup 1.

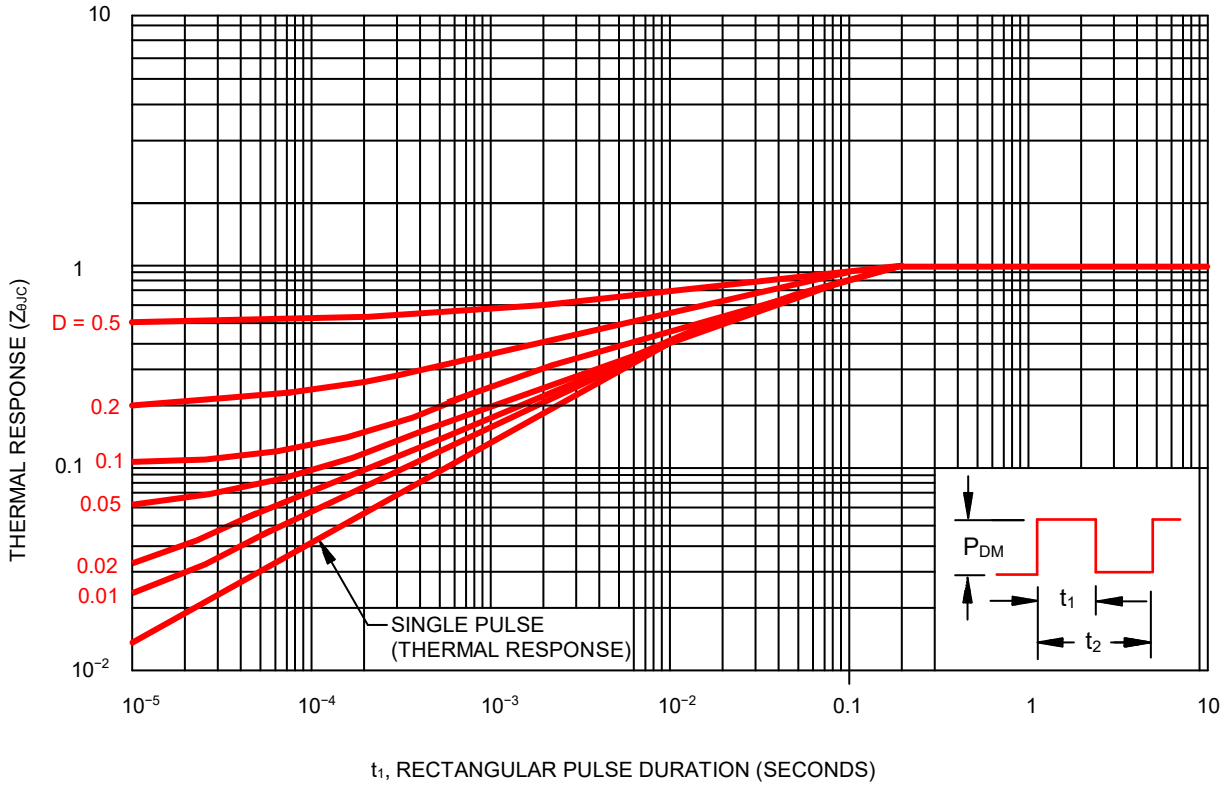
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TABLE II. Group E inspection (all quality levels) for qualification or re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Temperature cycle	1051	Condition G, 500 cycles.	
Hermetic seal	1071	Fine and gross leak.	
Electrical measurements		See table I, subgroup 2.	
Subgroup 2 1/			45 devices, c = 0
Steady-state reverse bias	1042	Condition A, 1,000 hours.	
Electrical measurements		See table I, subgroup 2.	
Steady-state gate bias	1042	Condition B, 1,000 hours.	
Electrical measurements		See table I, subgroup 2.	
Subgroup 4			sample size, N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			3 devices, c = 0
Barometric pressure (reduced) (2N6968 and 2N6969 only)	1001	Test condition C, $I_{(ISO)}$ = 0.25 mA (maximum), $V_{ISO} = V_{DS}$	
Subgroup 10			
Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors	3476	Test conditions shall be derived by the manufacturer.	22 devices, c = 0

1/ A separate sample for each test may be pulled.

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NOTES:
1. Duty factor D = t_1 / t_2
2. Per unit base $R_{\theta JC}$ = 1.8 °C/W
3. $T_{JM} = P_T * R_{\theta JC} * r(t) + T_C$

FIGURE 2. Thermal response curves.

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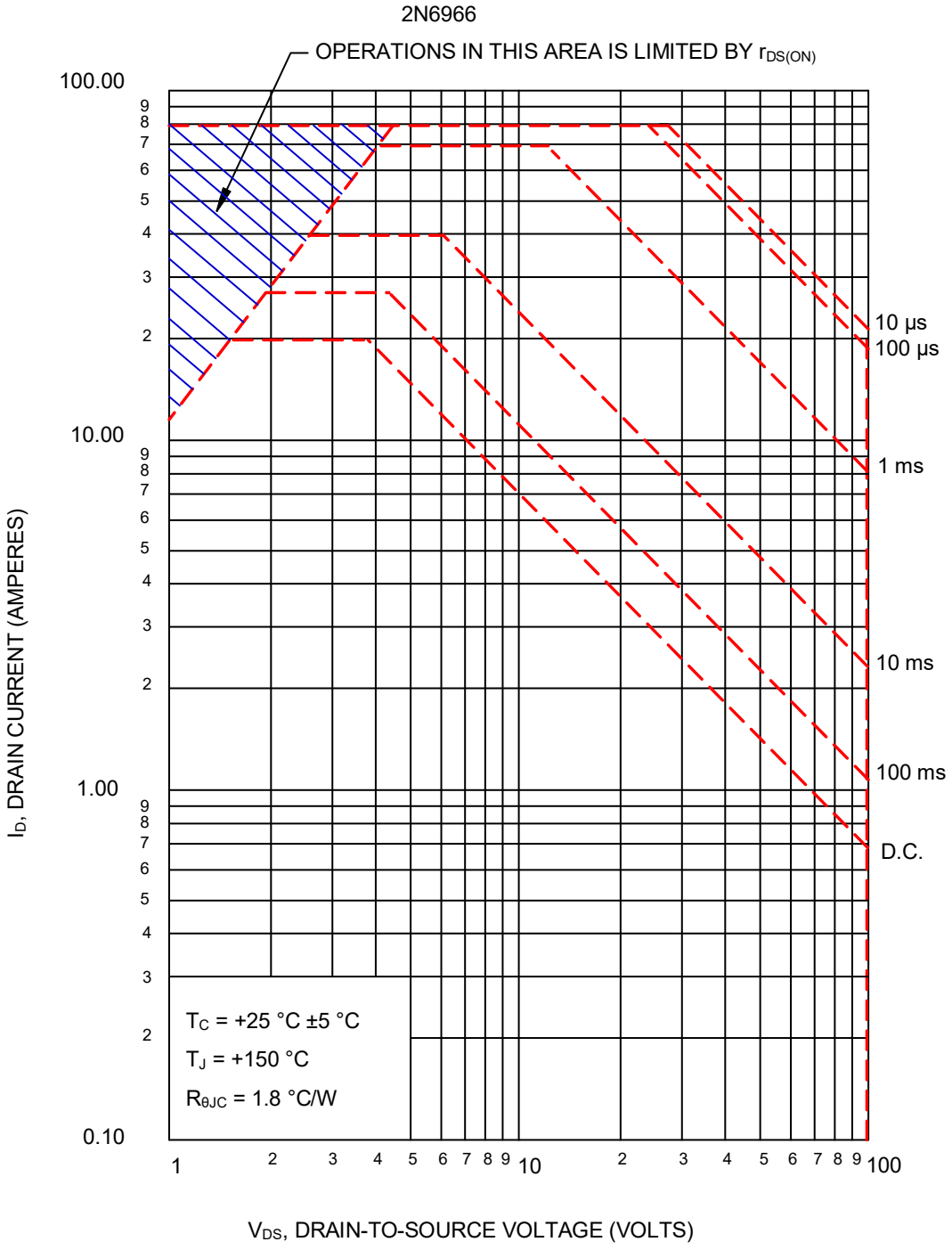


FIGURE 3. Safe operating area.

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2N6967

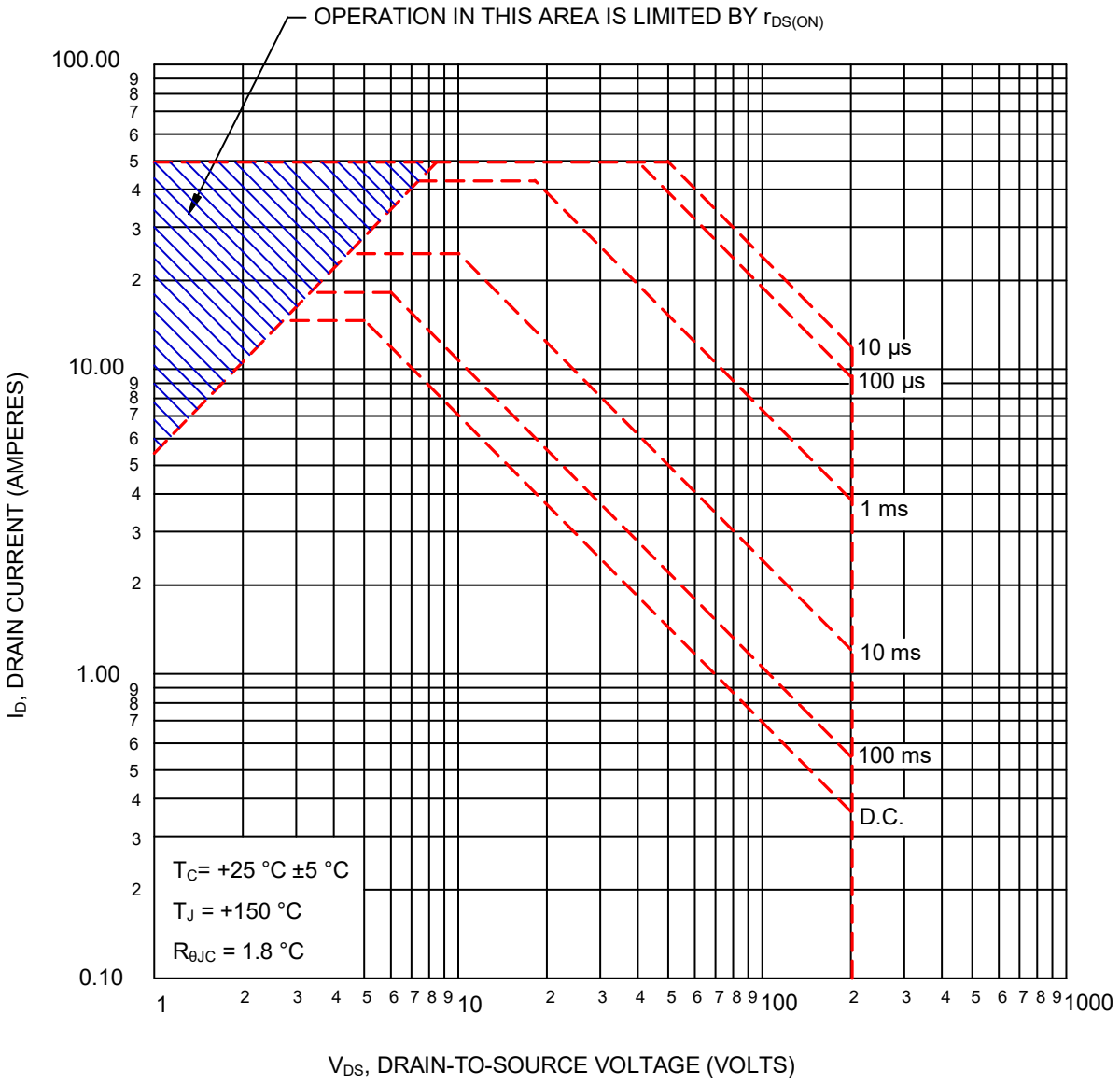


FIGURE 3. Safe operating area - Continued.

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2N6968

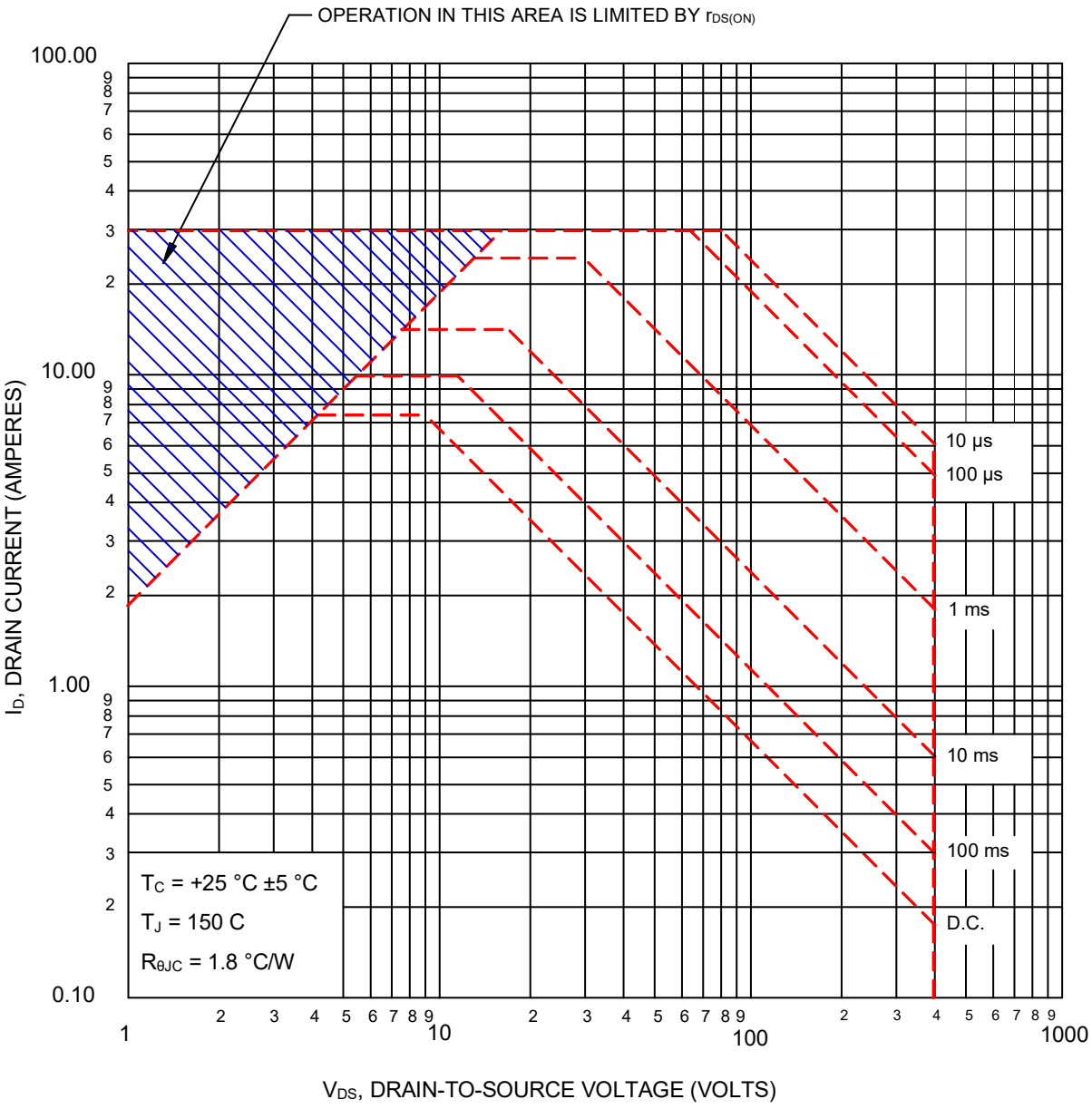


FIGURE 3. Safe operating area - Continued.

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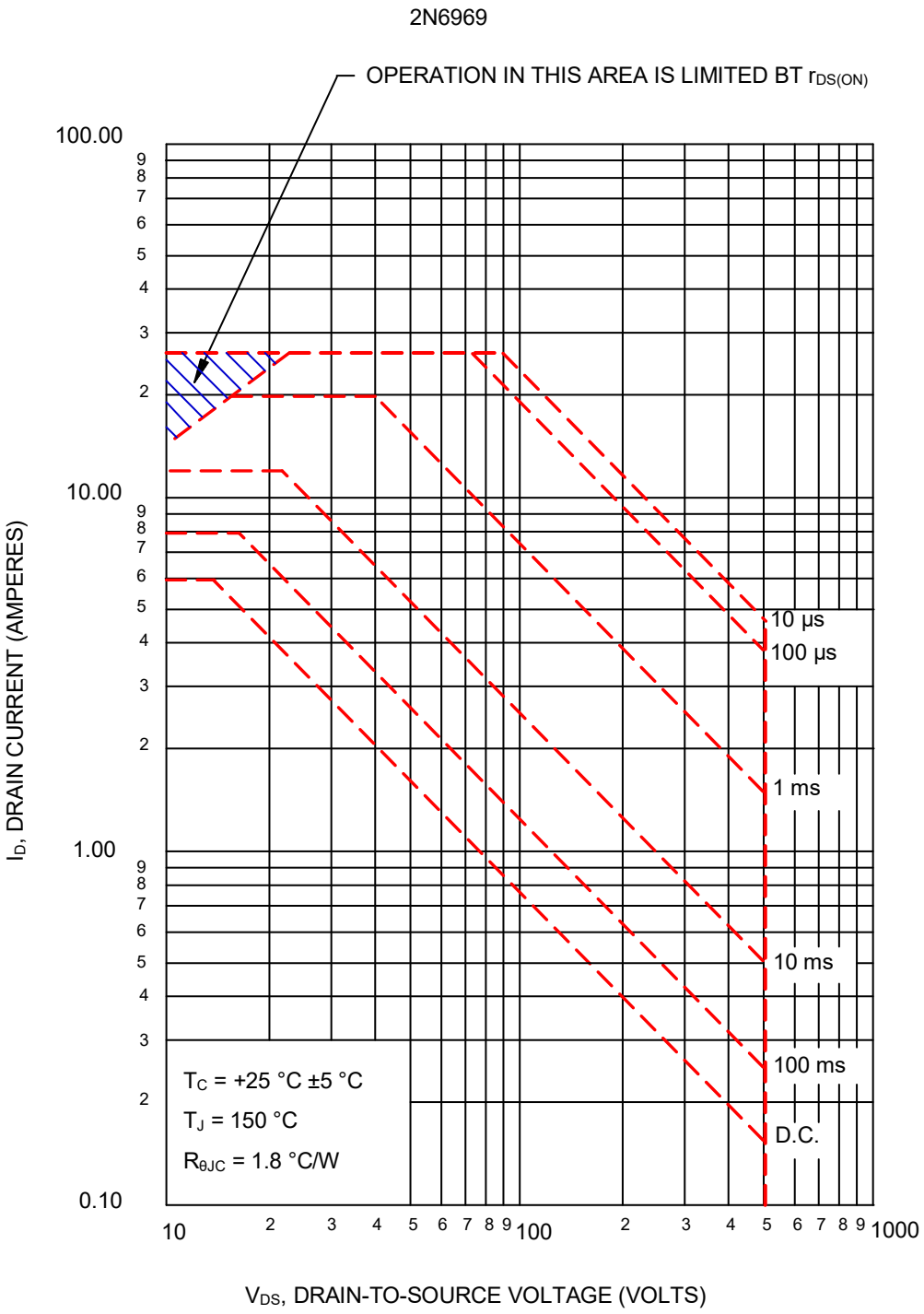


FIGURE 3. Safe operating area - Continued.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

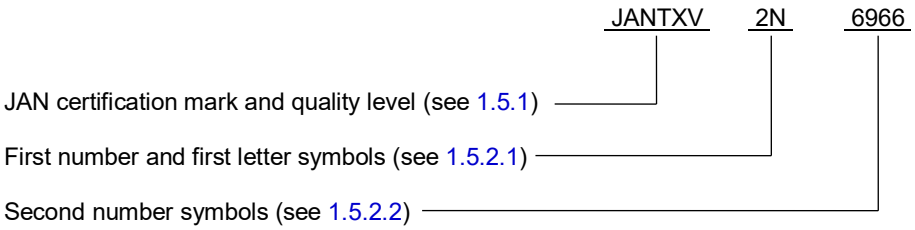
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.



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6.5 List of PINs. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality level
JAN2N6966	JANTX2N6966	JANTXV2N6966	JANS2N6966
JAN2N6967	JANTX2N6967	JANTXV2N6967	JANS2N6967
JAN2N6968	JANTX2N6968	JANTXV2N6968	JANS2N6968
JAN2N6969	JANTX2N6969	JANTXV2N6969	JANS2N6969

6.6 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's PIN. This information in no way implies that manufacturers' PIN's are suitable as a substitute for the military PIN.

PIN	Commercial types
2N6966	IRFJ140
2N6967	IRFJ240
2N6968	IRFJ340
2N6969	IRFJ440

6.7 Changes from previous issue. The margins of this specification are not marked with asterisks due to the extent of the changes.

Custodians:
Army − CR
Navy − EC
Air Force − 85
NASA − NA
DLA − CC

Review activities:
Army − AR, SM
Navy − AS, CG, MC, OS, SH
Air Force − 19, 70, 170

Preparing activity:
DLA − CC

(Project 5961−2020−081)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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